Supplementary Table 3.1. Results from Monte Carlo simulations comparing asymptotic coverage probabilities on the target causal parameter from the four existing MR methods and the proposed method: Scenario's A and B as a function of instrument strength.

Instrument Strength (α_1)	MR Method	Scenario A		Scenario B	
		Num Converged	Coverage Probability Estimate	Num Converged	Coverage Probability Estimate
0.01	Wald	7991	100.00%	7990	97.50%
	2SPS	7991	95.10%	7990	96.10%
	2SRI	8000	94.80%	8000	95.10%
	GMM	3809	100.00%	6482	99.90%
	IV-MVB	7946	100.00%	7919	100.00%
0.1	Wald	7986	100.00%	7988	95.50%
	2SPS	7986	95.40%	7988	96.10%
	2SRI	8000	95.10%	8000	94.90%
	GMM	4195	100.00%	6564	99.90%
	IV-MVB	7929	100.00%	7907	99.90%
0.2	Wald	7996	99.90%	7997	89.00%
	2SPS	7996	95.30%	7997	96.10%
	2SRI	8000	95.10%	8000	94.80%
	GMM	5091	100.00%	6793	100.00%
	IV-MVB	7959	99.90%	7927	99.90%
0.3	Wald	7998	99.50%	7999	76.80%
	2SPS	7998	94.70%	7999	95.80%
	2SRI	8000	94.50%	8000	94.70%
	GMM	6180	100.00%	7151	100.00%
	IV-MVB	7984	99.50%	7974	99.70%
0.5	Wald	8000	98.80%	8000	45.20%
	2SPS	8000	94.70%	8000	95.80%

Instrument Strength (α_1)	MR Method	Scenario A		Scenario B	
		Num Converged	Coverage Probability Estimate	Num Converged	Coverage Probability Estimate
	2SRI	8000	94.50%	8000	94.60%
	GMM	7523	100.00%	7635	100.00%
	IV-MVB	7998	98.70%	7998	98.80%
0.7	Wald	8000	98.00%	8000	18.40%
	2SPS	8000	95.30%	8000	96.20%
	2SRI	8000	95.20%	8000	95.00%
	GMM	7922	99.70%	7907	99.80%
	IV-MVB	8000	97.90%	8000	97.90%
1.0	Wald	8000	96.50%	8000	2.90%
	2SPS	8000	95.10%	8000	95.40%
	2SRI	8000	95.00%	8000	94.30%
	GMM	7997	98.60%	7996	98.60%
	IV-MVB	8000	96.40%	8000	95.90%

Supplementary Table 3.2. Results from Monte Carlo simulations comparing asymptotic coverage probabilities on the target causal parameter from the four existing MR methods and the proposed method: Scenario's C and D as a function of sample size.

Sample Size	Method	Scenario C		Scenario D	
		Num Converged	Coverage Probability Estimate	Num Converged	Coverage Probability Estimate
50	Wald	7878	99.30%	7890	88.60%
	2SPS	7878	96.80%	7890	97.90%
	2SRI	8000	96.50%	8000	97.00%
	GMM	6372	100.00%	6697	99.60%
	IV-MVB	7384	99.90%	6793	99.90%
200	Wald	7992	99.80%	7994	74.50%
	2SPS	7992	95.10%	7994	96.00%
	2SRI	8000	94.90%	8000	95.00%
	GMM	6705	100.00%	7018	100.00%
	IV-MVB	7982	99.60%	7960	99.70%
500	Wald	8000	98.90%	8000	45.50%
	2SPS	8000	94.70%	8000	95.90%
	2SRI	8000	94.50%	8000	94.80%
	GMM	7510	100.00%	7606	100.00%
	IV-MVB	7999	98.80%	7998	98.90%
1000	Wald	8000	97.90%	8000	18.60%
	2SPS	8000	95.00%	8000	95.50%
	2SRI	8000	94.80%	8000	94.50%
	GMM	7924	99.80%	7896	99.70%
	IV-MVB	8000	97.60%	8000	97.50%
1500	Wald	8000	97.20%	8000	7.10%
	2SPS	8000	95.30%	8000	96.00%

Sample Size	Method	Scenario C		Scenario D	
		Num Converged	Coverage Probability Estimate	Num Converged	Coverage Probability Estimate
	2SRI	8000	95.20%	8000	94.90%
	GMM	7985	99.50%	7978	99.40%
	IV-MVB	8000	97.20%	8000	97.10%
2000	Wald	8000	96.40%	8000	2.70%
	2SPS	8000	94.80%	8000	95.90%
	2SRI	8000	94.70%	8000	94.70%
	GMM	7999	98.70%	7998	98.80%
	IV-MVB	8000	96.30%	8000	96.50%
2500	Wald	8000	96.20%	8000	1.00%
	2SPS	8000	94.70%	8000	95.70%
	2SRI	8000	94.60%	8000	94.70%
	GMM	7997	98.20%	7997	97.90%
	IV-MVB	8000	96.10%	8000	96.00%